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Original Article

Improving Children's Mental Health with a Digital Social Skills Development Game: A Randomized Controlled Efficacy Trial of *Adventures aboard the S.S. GRIN*

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Abstract

Objective: The purpose of this study was to investigate whether a computer-based game to improve social skills and mental health in children with social skills deficits would be efficacious. The program, *Adventures aboard the S.S. GRIN*, translates a proven in-person intervention into a nine-episode interactive online adventure game that provides opportunity for knowledge acquisition and skill practice.
Materials and Methods: Participants (children aged 7–11 years with social skills challenges) were randomly assigned to immediate treatment group ($n = 33$) or waitlist control group ($n = 36$). Children in the immediate treatment condition completed the game at home over the course of 9 weeks. Before playing the game and again within 1 week of game completion, children completed surveys about social literacy, social anxiety, bullying, social self-efficacy, and social satisfaction.
Results: Children who played *Adventures* improved significantly more from pretest to posttest than children who did not play the game in social literacy, social anxiety, bullying victimization, and social satisfaction.
Conclusion: Online interactive games can be effective in improving mental health for children who struggle with social skills. For children who can access them, serious games have the potential to increase the reach of effective programs by overcoming the logistical and implementation barriers (such as cost, travel, and accessibility) that limit traditionally delivered mental health interventions.

Keywords: Intervention, Mental health, Serious games, Social skills

Introduction

The link between mental health and social and emotional skills

An estimated 15 million children suffer from a diagnosable mental health disorder, and many more are at risk of developing one.[1] The costs to individuals, families, and society of treating mental health disorders are substantial.[2,3] Fortunately, mental, emotional, and behavioral disorders can be prevented and treated by promoting social–emotional skills.[3] Social skills (factors such as interpersonal behaviors, peer acceptance, and communication that predicts social outcomes) and emotional skills (e.g., recognizing emotions in oneself and others and managing emotions) are closely intertwined and often referred to singularly as social–emotional skills.[4,5]

Unaddressed problems with social–emotional skills in childhood become risk factors for more chronic and severe mental health problems in adolescence,[6,7] but quality interventions can enhance these skills and circumvent poor outcomes.[8–10] Programs that focus on children who are showing indications of social, emotional, or behavioral problems are important from a public health and policy perspective; addressing issues early, before they require intensive mental health treatment, can save time and money in later, more intensive intervention.[9]

The problem of unmet need

Despite the importance of social–emotional skills for mental health and the proven effectiveness of quality intervention, it is estimated that only one-third of youth in need of mental health services actually receive them[11] because a number of barriers to treatment exist. Social–emotional skills intervention is typically delivered in person by a trained

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mental health professional. This can be costly, and there are not enough trained providers to meet the need for intervention. Constraints on schedules and location add to the burden for many families, and some children and families feel a stigma around receiving mental health services in a public setting. Other factors such as using an ineffective intervention or straying from program intention because of inadequate training, lack of staff time, instructional drift, or inadequate tools to support ongoing intervention can increase intervention variability and result in less-effective intervention.[8,9,12] We need a way to decrease the cost and increase the reach of programs that are proven to be effective.

The digital games solution

An innovative approach to this challenge is to create digital game interventions that deliver program content in accessible, engaging ways so that evidence-based interventions can be delivered consistently to many more children. A growing body of research demonstrates that online interventions can be effective in improving mental health.[13–15] Although it is important to acknowledge the digital divide that limits online access to children with lower incomes, digital games are accessible by children with device and Internet access and use. For the 80% of Americans who have Internet access in the home,[16] digital games can be provided at a much lower cost than traditional intervention, and the standardized administration greatly reduces the variability that can dilute effectiveness. They can be dynamic and individualized, so that much like an in-person tutor, they respond to choices by selecting the most appropriate content for each child. Accessing the technology at home at convenient times reduces stigma, travel, and scheduling barriers.

The benefits of technology game-based interventions extend beyond addressing barriers to traditional didactic approaches. A basic tenet of developmental psychology is that play is a fundamental way in which children learn. Playing games allows children to experiment with different social experiences and emotional consequences,[17] reproduce real-life conflicts in order to work out ideal resolutions,[18] develop their understanding of social situations and their role in them,[19] and achieve emotional mastery.[20] These basic functions of games have also been tied specifically to computer games[21] and are reinforced by the engaging and appealing nature of digital gameplay.[22–24]

Digital games can immerse the player in decision-making situations, provide a safe environment to practice emerging skills without concerns of embarrassment or social consequences, and lead to higher confidence when applying the learning in real life.[25] Granic et al.[21] postulate that ''the pretend context of videogames may be real enough to make the accomplishment of goals matter but also safe enough to practice controlling, or modulating, negative emotions in the service of these goals'' (p. 72).

This assertion is borne out in research evidence, which shows that skills learned in videogame play transfer to real world knowledge and behavior change in a number of areas, including social skills and mental health.[26,27] Other studies have shown that playing videogames that reward prosocial behavior are predictive of increased prosocial behavior in real life both immediately and in the long term.[28,29] Granic et al.,[21] after reviewing the literature on the benefits of playing videogames, emphasized the potential of videogames for mental health interventions for youth, stating that ''Developing therapeutic video games can address (the gap between what youth actually know and what they do in their everyday lives) because they engage players in immersive emotional experiences …and provide opportunities to practice new regulatory skills until they are automatized, canalizing new neural patterns that can lead to generalization outside the game'' (p.75). An early pilot test of the first half of the *Adventures* program provides additional support for this idea by demonstrating that children who played the program at home showed greater gains in real world social skills and behaviors than children who did not.[30]

The present study

In this article, we report findings from an efficacy study of a digital social skills game, *Adventures aboard the S.S. GRIN*. The game is designed for children aged 7–11 who struggle with social skills. An earlier test of a beta version of this game that included the first four episodes was well received by parents and children.[30] In addition, parents reported lower psychosocial distress and greater behavioral and emotional strength at posttest for children who had played compared to those who had not. The purpose of the present study was to test whether children with social skills challenges report better social–emotional skills knowledge and improved mental health outcomes as a result of playing *Adventures*.

Materials and Methods

Participants

Participants were recruited in the fall of 2014 through postings on parenting listservs, social media advertisements targeted to parents of children aged 7–11 years, and through contacts with area and national school systems and child service providers. Recruitment materials included a brief description of the Adventures program and details about study activities and eligibility requirements. Interested parents completed an online eligibility survey, which included demographic information and five subscales of the Behavior Assessment System for Children, Second Edition (Adaptability, Aggression, Atypicality, Conduct Problems, Withdrawal; BASC-2).[31] Child eligibility requirements were (1) being between the ages of 7 and 11 years in September 2014, (2) receiving instruction in a regular education classroom for at least 40% of the school day, (3) English language proficiency, (4) access to a computer with Internet use at home, and (5) scoring at or above the clinical cutoff for their age and sex on any of the 5 BASC-2 subscales, or in the ''at-risk'' range on at least two subscales.

The BASC-2 is a norm-referenced screening system for measuring behavioral and emotional strengths and weaknesses. To identify children struggling with social–emotional skills, we included the subscales relevant to those skills; to reduce participant burden, we did not include BASC-2 subscales less directly related to the social–emotional domain (e.g., study skills, activities of daily living). Fourteen children whose parents completed the eligibility survey did not meet the criteria for enrollment; five were excluded based on age, three because they were not in a regular education setting at least 40% of the school day, and six because they did not meet BASC-2 inclusion criteria.

Table 1. Comparison of Demographics and Behavior Assessment System for Children, Second Edition Scores by Condition at Baseline

	Treatment n = *33 (%)*	*Control* n = *36 (%)*	χ^2	P
Demographics				
Sex: M/F	58/42	61/39	0.089	0.765
Race: White/Non-White	55/45	56/44	0.007	0.933
Ethnicity: Hispanic/Non-Hispanic	6/94	6/94	0.008	0.929
Age: 7/8/9/10/11	15/21/37/9/18	14/22/36/19/9	2.581	0.630
BASC-2				
Adaptability: average/at-risk/clinical	58/12/30	61/11/28	0.089	0.956
Aggression: average/at-risk/clinical	55/33/12	47/31/22	1.234	0.540
Atypicality: average/at-risk/clinical	36/24/40	33/20/47	0.470	0.790
Conduct: average/at-risk/clinical	61/33/6	61/22/17	2.443	0.295
Withdrawn: average/at-risk/clinical	18/42/40	14/33/53	1.242	0.537

BASC-2, Behavior Assessment System for Children, Second Edition.

To reduce sampling error, eligible participants were stratified by age, sex, race, ethnicity, and BASC-2 subscale scores. Following stratification, participants were randomly assigned to either an immediate treatment condition or a wait-list control condition that got access to *Adventures* after posttest data collection was complete. Study attrition (i.e., failure to complete all nine episodes and the pre- and posttest assessments) was 26% ($n = 24$) over the entire data collection period; there were no significant group differences in attrition, as determined by Chi-square tests. Descriptive information for participants is presented in Table 1, separately for participants in the treatment and control groups. Chi-square tests revealed no significant differences at baseline between participants in the two study conditions on the categorical variables age, sex, race, ethnicity, or BASC-2 subscale scores. In addition, *t*-tests on the scale-based outcome variables (described below) revealed no significant differences at pretest between participants in the treatment and control conditions, or between participants who attrited and those who completed the study.

Prototype development and refinement

Details of the game are shown in Table 2. *Adventures* was created by translating the skills taught in an evidence-based, in-person social skills program into a game that teaches the skills in an engaging narrative-driven context and responds intelligently to children's choices in the program. Children's experience in the game varies based on their previous performance. The game provides feedback, hints, and prompts based on performance and requires skill mastery for progression. The program we developed into *Adventures* is *Social Skills Group Intervention* (SSGRIN), an in-person intervention that has been used by thousands of students for over 20 years, and results in significant improvements in social–emotional skills and mental health.[32–34]

To develop *Adventures*, we translated the SSGRIN curriculum into a digital game by designing in-game tasks that require children to apply specific social–emotional skills to solve social problems encountered in the game. Each social skill taught in Adventures involved a series of in-game interactions, in which the participant made choices to learn and practice social skills. For example, to train participants on social initiation, we presented three scenarios that are true-to-life situations for elementary school children: approaching an individual nonplayer character (NPC) that appears easy to talk to, joining a group of NPCs at a game in progress, and approaching small groups of NPCs that appear less easy to talk to (e.g., they look older and ''cooler'' than the initial NPC contact). Within each scenario, the player had choices such as approaching or avoiding; interrupting or waiting; and being passive, assertive, or aggressive in initiating interaction and responding to NPC reactions. Feedback to choices and prompts to respond were presented at appropriate times throughout the program.

An interdisciplinary team of psychologists, computer programmers, and artists collaborated to create an engaging learning environment to serve as the context for social problem solving (team members and roles shown in Table 3). We chose a single-player format for *Adventures* to enable individualized feedback and play paths based on player choices, allow for control of the reactions of other characters in a game, create a safe environment in which to practice fledgling skills without social ramifications, and avoid the possible iatrogenic effects of participants reinforcing negative behaviors in other children. As children navigate the game, the software captures in-game behaviors (such as the initiation behaviors described above) and uses these data in real time to assess the child's progress and determine the appropriate next step. Some of these behaviors are dialog-driven menu choices, whereas others are game-based behavior. All text-based dialogue is accompanied by audio voiceover.

The inclusion of appealing narrative and challenges was key in the development of the game, as these aspects are critical to engage children in instructional software.[35,36] Players begin by creating a customized avatar to navigate the digital environment, interact with NPCs, and solve in-game challenges. The game has a nautical adventure theme in which the player is a new recruit on the sailing ship S.S. GRIN and joins an established crew, each of whom specializes in a specific social skill. The crew members are the child's companions in the narrative and serve as pedagogical agents. The child's avatar joins the crew and travels around an island, interacting with characters and solving social challenges to address plot conflicts and save friendship on

Table 2. Characteristics of a Videogame for Mental Health: Adventures Aboard the S.S. GRIN

Characteristics	*Description*
Overview	
Health topic	Social and emotional skill development
Targeted age group	Children aged 7–11 years
Other targeted group characteristics	Children with social/emotional skills deficits
Short description of game idea	Single-player point-and-click adventure game that teaches social skills as player solves in-game social challenges
Target player	Individual
Guiding knowledge or behavior change theory, model, or conceptual framework	Evidence-based social skills training SSGRIN
Intended health behavior changes	Improved social skills
Knowledge elements to be learned	Respect, consequences, responsibility, impulse control, communication, perspective taking, friendship skills, social initiation, cooperation, compromise, emotion regulation
Behavior change procedure (taken from Michie inventory) or therapeutic procedures employed	Behavioral rehearsal/practice, covert learning, feedback
Clinical or parental support needed?	No. Optional supplemental activities can be done with adult participation
Data shared with parent or clinician	Yes
Type of game	Educational
Story (if any)	
Synopsis including story arc	The child creates a customized avatar to represent himself/herself in the game. The player is a new recruit on the sailing ship S.S. GRIN and joins an established crew, each of whom specializes in a specific social skill (e.g., impulse control). The narrative purpose of the crew members is to be the child's companions, and the game also employs them as pedagogical agents to provide feedback and assistance as needed. The player navigates the virtual environment, interacting directly with NPCs and learning to appropriately manage social challenges as the crew travels around an island addressing conflicts and saving friendship on the island.
How the story relates to targeted behavior change	The social challenges woven into the plot require the player to learn and use appropriate social skills.
Game components	
Player's game goal/objective	The narrative objective is to gather seven missing friendship stones and restore them to their rightful place; the education objective is to successfully complete a series of social problem-solving scenes of increasing complexity to demonstrate skill mastery
Rules	Player must make best social/emotional responses (menu- or action-based) to achieve badges and complete each scene
Game mechanics	Single-player point-and-click; player navigates environment and makes behavioral and menu choices by clicking mouse
Procedures to generalize or transfer what's learned in the game to outside the game	NPCs review the social and emotional skills learned in the game; quizzes require the player to think about skill application in the real world; optional supplemental activities provide extra practice of the skills in a real world setting
Virtual environment	
Setting	2.5D virtual world consists of a tropical island setting with 10 different settings around the island and on a ship
Avatar	
Characteristics	Customizable (gender, skin color and style, hair color and style, clothing color and style)
Abilities	Movement in 2.5D environment
Game platforms needed to play the game	Computer
Sensors used	Not applicable
Estimated time to play	Approximately 3.75 hours in total (nine episode of ~25 minutes each)

NPCs, nonplayer characters; SSGRIN, *Adventures* is *Social Skills Group Intervention*.

Table 3. Game Development Roles

Role	*Individual(s)*
Principal Investigators	Rebecca Sanchez, Melissa DeRosier
Lead Game Designer	Chris Hehman
Instructional Content	Rebecca Sanchez
Research	Rebecca Sanchez, Chelsea Bartel, Emily Brown
Art	Jeremiah Weatherley, Chrystin Garland, Kim Smith
Lead Audio Engineer	Wes Sommer
Systems Architecture and Tools/Lead Programmer	Matt Habel
Quality Assurance	James Dooley, Jennifer Hesse

the island. Lines of mystery and ''cliff-hangers'' are woven throughout the story to maintain engagement across episodes. Screenshots from *Adventures* are shown in Figure 1.

Measures

Social literacy. The Achieved Learning Questionnaire (ALQ), developed by 3C Institute, is a 20-item measure of children's social literacy across the eight domains covered by the Adventures program (respect, impulse control, communication, social planning, empathy, social initiation, cooperation/compromise, and social initiation). Children were asked to demonstrate their understanding of social skills learned throughout the *Adventures* curriculum by answering a variety of multiple choice questions, such as, ''Which of these is the best example of acting cooperatively?'' The number of correct responses was calculated for the total score with higher numbers indicating greater social skills knowledge. No psychometrics are available for this investigator-created measure of social skills knowledge.

Social self-efficacy. Social self-efficacy, or children's confidence in their ability to succeed socially, was assessed using the 10-item self-efficacy subscale of the Self-Efficacy and Outcome Expectancy Measure.[37] Children rate how sure they are, on a 5-point scale ranging from ''Not at all'' (1) to ''Really sure'' (5), that they could perform a variety of social behaviors. Representative items of the scale include ''How sure are you that you could start talking with a kid your age who you just met?'' and ''When a kid your age does a good job at something, how sure are you that you could tell them they did a good job?'' The mean across items was used for analysis, with higher scores indicating greater social self-efficacy. This measure has high internal consistency ($\alpha = 0.87$).[37]

Social anxiety. The Social Anxiety Scale for Children—Revised[38]—is an 18-item measure of children's worries about various social situations with same-aged peers. Children indicate how much they agree with statements such as ''I worry about being teased'' and ''It's hard for me to ask other kids to do things with me'' on a 5-point scale ranging from ''Not at all'' (1) to ''All the time'' (5). The mean across items were used for analysis, with a higher score being indicative of greater anxiety. The Social Anxiety Scale for Children has demonstrated acceptable internal consistency ($\alpha = 0.65$).[38]

Social satisfaction. The Loneliness and Social Dissatisfaction scale[39,40] is a 19-item measure on which children rate their satisfaction with peer relationships. Children indicate on a 4-point scale from ''Not at all'' (1) to ''Almost Always'' (4) how true each statement and situation is for them. Representative items of this measure include ''It is easy for me to make friends'' and ''I have kids to talk to.'' The mean across items was used for analysis, with higher scores representing higher satisfaction with peer relationships. The Loneliness and Social Dissatisfaction Scale has high reliability ($\alpha = 0.91$) and high internal consistency ($\alpha = 0.90$).[39]

Bullying victimization. The Personal Experiences Checklist (PECK)[41] assesses bullying victimization. Children rate how often bullying events such as ''Other kids punch me'' and ''The other kids ignore me on purpose'' happened to them in the last month on a 5-point scale ranging from ''Never'' (1) to ''Every Day'' (5). The mean across items were used for analysis, with higher scores indicating greater experiences of bullying victimization. The PECK has been shown to have good internal consistency (Cronbach's $\alpha = 0.79$).[41]

Bullying perpetration. The Bullying Others Subscale of the California Bullying Victimization Scale[42] was used to assess bullying perpetration. Children report how often they have engaged in bullying behaviors such as ''Teased or called another child names in a mean or hurtful way'' and ''Stolen or damaged another child's things in a mean or hurtful way'' in the past month on a 5-point scale from ''Not in the past month'' (1) to ''Several times a week'' (5). The mean across items was used for analysis with higher scores indicating more bullying behaviors. Psychometric properties for this measure are not yet available as it is still being piloted by the authors.

Procedure

The study protocol was approved by the 3C Institute Institutional Review Board (00002238) before data collection began. Informed parent permission and child assent were obtained from participants before participation. All activities were completed online through a secure project website. Surveys were delivered through an interactive quiz-show program wherein an animated character read each question and the answer choices aloud. The character could be muted if children preferred to read the questions themselves. Parents were instructed to allow their child to complete surveys independently. Survey completion took about 30 minutes at each time point (pre and post). The week following pretest data collection, the treatment group began playing *Adventures*. Episodes were released weekly for 9 weeks, and children had 1 week to complete each episode. Participants in the waitlist control group did not complete any study-related tasks during this time. Following the treatment group's 9-week interaction with the game, children in both groups completed measures again. Once posttest data

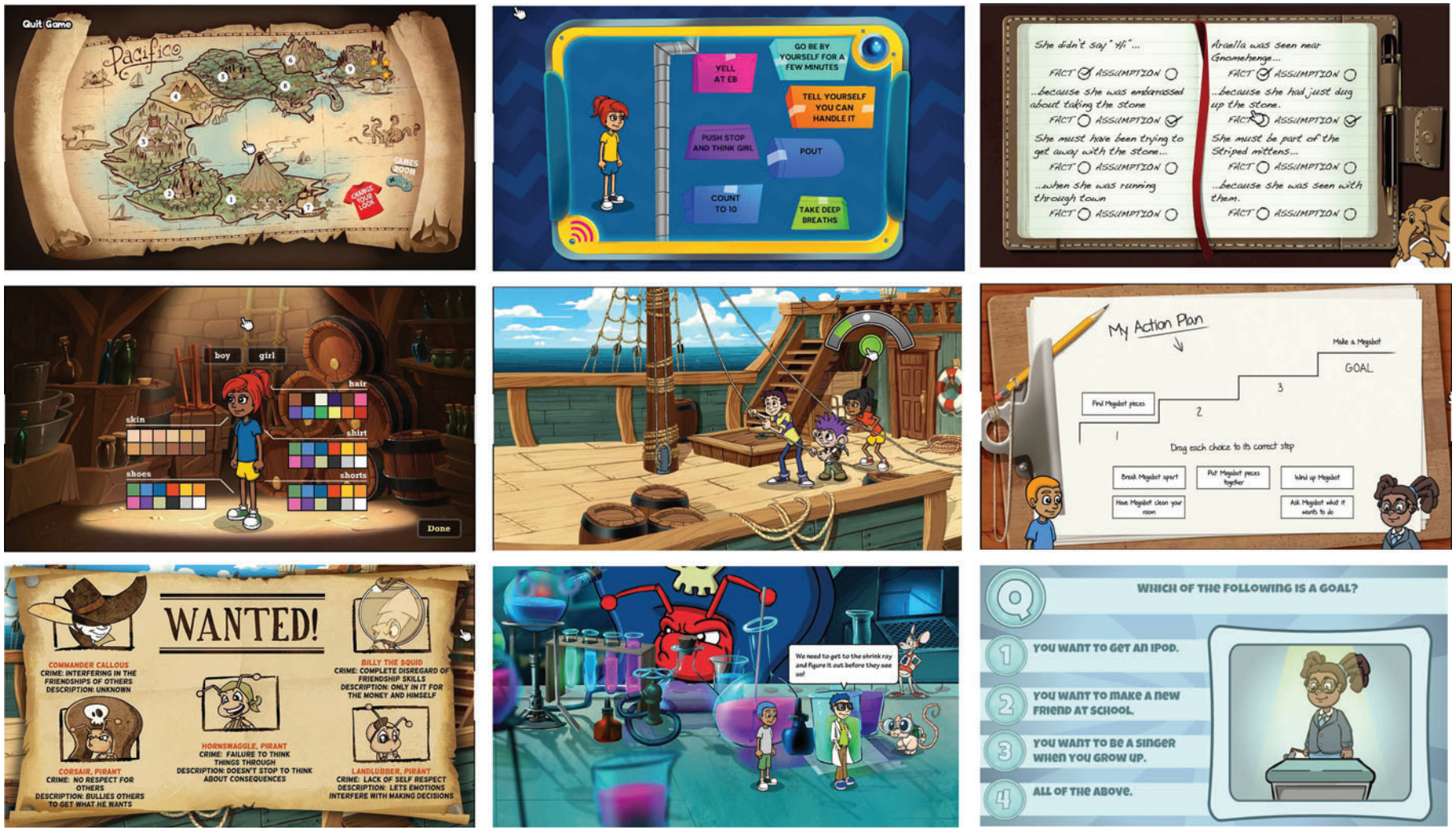

FIG. 1. Screenshots from Adventures aboard the S.S. GRIN. (Color images available at www.liebertonline.com/g4h)

Table 4. Summary of Relevant Descriptive Information, Pre–Post Change Comparisons, and Time by Condition Interaction Analysis of Variance Statistics and Effect Sizes

	Mean (SE)		*Pre–post change*		*ANOVA*		
	Preintervention	*Postintervention*	*Mean difference (SE)*	P	*F*	P	η_p^2
Social literacy							
Treatment	10.67 (0.58)	14.46 (0.56)	3.79 (0.53)	0.000	21.057	0.000	0.24
Control	10.75 (0.55)	11.19 (0.53)	0.444 (0.50)	0.381			
Social self-efficacy							
Treatment	3.41 (0.09)	3.58 (0.10)	0.173 (0.10)	0.083	2.550	0.115	0.04
Control	3.33 (0.09)	3.29 (0.10)	–0.044 (0.09)	0.094			
Social satisfaction							
Treatment	2.85 (0.09)	3.01 (0.09)	0.167 (0.07)	0.016	4.451	0.039	0.06
Control	2.89 (0.08)	2.86 (0.08)	–0.029 (0.06)	0.649			
Social anxiety							
Treatment	2.79 (0.10)	2.49 (0.90)	–0.304 (0.08)	0.000	4.110	0.047	0.06
Control	2.78 (0.09)	2.70 (0.09)	0.078 (0.08)	0.318			
Bullying victimization							
Treatment	2.15 (0.15)	1.72 (0.12)	–0.426 (0.08)	0.000	12.066	0.001	0.15
Control	2.16 (0.14)	2.11 (0.12)	–0.052 (0.08)	0.491			
Bullying perpetration							
Treatment	1.50 (0.09)	1.37 (0.09)	–0.894 (0.67)	0.184	2.799	0.099	0.04
Control	1.46 (0.09)	1.55 (0.08)	0.647 (0.64)	0.313			

ANOVA, analysis of variance; SE, standard error.

collection was completed, the wait-list control group received *Adventures* to interact with as they wished.

We hypothesized that children who played Adventures would demonstrate significantly greater improvements at posttest in social literacy, social self-efficacy, social satisfaction, social anxiety, bullying victimization, and bullying perpetration compared to children in the control group. To test these hypotheses, we conducted mixed-method analyses of variance for each outcome measure with the treatment group as the between-subjects variable and time as the within-subjects variable. Bonferroni adjustments were used to control for multiple comparisons.

Results

Results, including relevant statistics and effect sizes for these analyses, are shown in Table 4. Analyses revealed support for all but two of our a priori hypotheses with significant time by condition interaction effects found for social literacy, social satisfaction, social anxiety, and bullying victimization. The group by time interaction for social self-efficacy and bullying perpetration approached but did not reach statistical significance. Partial eta-squared values of 0.24 and 0.15 for social literacy and bullying victimization, respectively, represent large effects, whereas the social satisfaction and social anxiety findings represented a medium effect ($\eta_p^2 = 0.06$, for both).[43]

To further examine the significant interaction effects, we ran post hoc condition by time comparisons (Table 4), which revealed that children in the treatment condition demonstrated significant improvement from pretest to posttest on social literacy, social satisfaction, social anxiety, and bullying victimization, whereas children in the control group did not.

Finally, we examined developmental and gender differences in intervention effects by conducting separate multivariate analyses of variance with age and gender as the independent variables and treatment group pretest-to-posttest change scores on all outcome variables as the dependent variables. We found no significant differences based on age or gender on any outcome variable.

Discussion

The purpose of the present study was to investigate the efficacy of a new digital game designed to improve mental health for children struggling with social skills. We predicted that playing the game would lead to improvements in social–emotional skills knowledge, feelings, and behaviors. The efficacy of *Adventures* in increasing social–emotional skills knowledge and understanding was supported by greater gains from pretest to posttest on an ALQ for children who played *Adventures* compared to those who did not. Improved social literacy is a good initial step in behavior change; it is a necessary but not sufficient component of performance. It is therefore also important that the measures of social anxiety, social satisfaction, and bullying victimization were also significantly improved after playing *Adventures*.

Playing *Adventures* did not lead to a statistically significant increase in children's belief in their ability to succeed in social situations. One possible explanation for this result is that the game does not significantly increase social self-efficacy. Alternatively, the game may increase social self-efficacy, but this result was not detected in this small study; a fully powered investigation with a larger sample size may reveal that interacting with the game does increase social self-efficacy, consistent with the findings from the in-person SSGRIN.[32] Another possibility is that the effect of *Adventures* on social self-efficacy will continue to strengthen over time as children practice the social–emotional skills they have learned, become more confident in their ability to

use them, and reap the social and mental health benefits of better social–emotional skills.

Finally, expected decreases in bullying perpetration as a result of playing *Adventures* were not borne out in the results. Baseline incidence of bullying perpetration was very low, leaving little room to detect decreases in this behavior.

Limitations and future directions

This randomized controlled trial of *Adventures aboard the S.S. GRIN* provides evidence for the efficacy of the game in improving the social–emotional skills knowledge, feelings, and behaviors important to good mental health. Efficacy tests are a first step in proving full intervention effectiveness, and future development and testing of *Adventures* will provide more information about its capabilities and shed more light on the utility of games for improving mental health outcomes. Because the present study was designed as an initial efficacy test of the game, a number of limitations should be considered. First, our sample size was relatively low. As a result, the statistical power was limited and we were restricted to fairly basic statistical methods and may not have been able to detect significant between-group differences (i.e., social self-efficacy). The next step in our research will be large-scale testing of the game; the larger sample size will enable us to examine possible mediators and moderators in the relation between gameplay and mental health outcomes. For example, is the game more appealing or effective for one type of social challenge? Does playing the game increase social self-confidence, which in turn gives children the courage to try the solutions they learned when faced with social challenges?

We will also collect longer term follow-up data to investigate whether more time will correspond to greater benefits. It is possible that some effects of *Adventures* will weaken as time from intervention delivery increases, but it is also possible that, as with the in-person SSGRIN program on which *Adventures* is based,[34] the effects reported in this article will get stronger over time as children gain practice and confidence, and as others begin to respond predictably to the child's improved social–emotional skills. Another interesting research design modification would be to replace the ''business as usual'' control group used in the present study with an active control group that performs a specific task, such as interacting with an academic videogame. An active control group can guard against placebo mechanisms contributing to between-group differences.[44]

Self-report data are the only way to assess the personal beliefs, emotions, and experiences we investigated in this study. In future investigations, we will supplement the self-report mental health data examined in the present study with behavior reported by adults in the child's life (e.g., parents or teachers) to determine the effect of *Adventures* on observable behavior change. Finally, future investigations of the game will include an examination of how *Adventures* compares to other social–emotional skills interventions, including the in-person SSGRIN program. This comparative study will include both outcome effectiveness and cost effectiveness, as we expect significant savings with this digital approach.

Conclusion

A wealth of evidence shows that good social–emotional skills are important in the development of positive mental health and that children who struggle with these skills can improve with effective training. Access to high-quality training has been limited, however, due to logistical and implementation barriers associated with traditional training. The results of the present study demonstrate that digital games can be an effective strategy for improving the mental health of children who struggle with social skills. This emerging methodology holds promise for overcoming a number of barriers associated with traditional interventions and expanding the reach of mental health interventions. Digital games for mental health can meet the needs of many more children at risk for poor mental health outcomes.

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Author Disclosure Statement

The authors are employees of 3C Institute, which may benefit financially from the sale of this game.

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